20. Show that

(i) $\tan(45^\circ + A) = \dfrac{1+\tan A}{1-\tan A}$ (ii) $\tan(45^\circ - A) = \dfrac{1-\tan A}{1+\tan A}$.

21. Prove that $\cot(A+B) = \dfrac{\cot A \cot B - 1}{\cot A + \cot B}$.

22. If $\tan x = \dfrac{n}{n+1}$ and $\tan y = \dfrac{1}{2n+1}$, find $\tan(x+y)$.

23. Prove that $\tan\left(\dfrac{\pi}{4}+\theta\right)\tan\left(\dfrac{3\pi}{4}+\theta\right) = -1$.

24. Find the values of $\tan(\alpha+\beta)$, given that $\cot\alpha = \dfrac{1}{2}, \alpha \in \left(\pi, \dfrac{3\pi}{2}\right)$ and $\sec\beta = -\dfrac{5}{3}, \beta \in \left(\dfrac{\pi}{2}, \pi\right)$.

25. If $\theta + \phi = \alpha$ and $\tan\theta = k\tan\phi$, then prove that $\sin(\theta-\phi) = \dfrac{k-1}{k+1}\sin\alpha$.

3.5.2 Multiple angle identities and submultiple angle identities

In 1831, Michael Faraday discovered that when a wire is passed near a magnet, a small electric current is produced in the wire. This property is used to generate electric current for houses, institutions and business establishments throughout the world. By rotating thousands of wires near large electromagnets, massive amount of electricity can be produced.

Voltage is a quantity that can be modeled by sinusoidal graphs and functions. To model electricity and other phenomena, trigonometric functions and identities involving multiple angles or sub multiple angles are used.

If A is an angle, then $2A, 3A, \ldots$ are called **multiple angles** of A and the angle $\dfrac{A}{2}, \dfrac{A}{3}, \ldots$.are called **sub-multiple angles** of A. Now we shall discuss the trigonometric ratio of multiple angles and sub-multiple angles and derive some identities.

Double Angle Identities

Let us take up the sum and difference identities and examine some of the consequences that come from them. Double angle identities are a special case of the sum identities. That is, when the two angles are equal, the sum identities are reduced to double angle identities. They are useful in solving trigonometric equations and also in the verification of trigonometric identities. Further double angle identities can be used to derive the reduction identities (power reducing identities). Also double angle identities are used to find maximum or minimum values of trigonometric expressions.

Identity 3.7: $\sin 2A = 2\sin A\cos A$

Proof.

We know that $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Taking $\alpha = \beta = A$, we have $\sin(A+A) = \sin A\cos A + \sin A\cos A$

Thus, $\sin 2A = 2\sin A\cos A$.

(i) $y = \sin 2x$ and $y = 2\sin x$ are different. Draw their graphs and identify the difference.

(ii) Application of $\sin 2A = 2\sin A\cos A$: When an object is projected with speed u at an angle α to the horizontal over level ground, the horizontal distance (Range) it travels before striking the ground is given by the formula $R = \dfrac{u^2\sin 2\alpha}{g}$.

Clearly maximum of R is $\dfrac{u^2}{g}$, when $\alpha = \dfrac{\pi}{4}$.

(iii) $|\sin A\cos A| = \left|\dfrac{\sin 2A}{2}\right| \le \dfrac{1}{2}$.